

Kiernan Devane

484-723-9238 | kiernantd17@outlook.com | [linkedin.com/in/kiernan-devane](https://www.linkedin.com/in/kiernan-devane) | github.com/kiernantd | kiernan-devane-portfolio.vercel.app

Education

University of Pittsburgh

Bachelor of Science in Computer Science, Minor in Linguistics

Pittsburgh, PA

Aug. 2022 – Dec. 2025

GPA: 3.3, Cum Laude | Relevant Coursework: Operating Systems, Data Structures & Algorithms, Applied Cryptography, Computer Organization, Software Quality Assurance

Experience

Surreality Lab

Junior Research Engineer

Pittsburgh, PA

Jan. 2025 – May 2025

- Diagnosed a low-light issue in captured AR frames and resolved it by introducing a startup delay before capture, allowing the camera sensor to complete auto-exposure calibration before the detection pipeline began processing.
- Architected a multithreaded producer/consumer pipeline decoupling AVFoundation frame capture from OpenCV-based ArUco marker detection on a background thread, evaluating Swift actors before landing on a mutex-based design for tighter frame-queue control.
- Collaborated in Agile sprint cycles with researchers and engineers to deliver production-ready AR tracking features for a surgical training application targeting the Apple Vision Pro.

Projects

Stampede — Scalable Event Booking Backend | *Python, FastAPI, PostgreSQL, Redis, Docker, AWS Cognito*

Feb. 2026 – Present

- Designing a high-concurrency REST API for seat reservations using Redis distributed locks and PostgreSQL partial unique indexes as a two-layer guarantee against double-booking race conditions across simultaneous requests.
- Architecting a relational schema with row-level locking and tuned transaction isolation to ensure booking consistency under contention; containerized with Docker and integrating AWS Cognito for stateless JWT-based authentication.
- Planning sliding-window rate limiting and Locust load-test benchmarks in upcoming phases to validate throughput targets and surface bottlenecks across the full reservation and confirmation lifecycle.

TUI Personal Website | *TypeScript, Ink (React for CLIs), Docker, Fly.io*

May 2026

`ssh -p 2222 guest@kiernan-devane-personal.fly.dev`

- Built a terminal UI personal site in TypeScript using Ink (React for CLIs) and Figlet for ASCII rendering, fully navigable via keyboard with dynamic routing between sections.
- Architected deployment with the Node.js app serving as the OS-level SSH login shell, eliminating any web server layer; containerized with Docker and deployed on Fly.io with a dedicated TCP port 2222 endpoint.

Secure Cryptographic Messaging System | *Python, AES, RSA, SHA-512* Sep. 2024 – Dec. 2024

- Built with a team of five in an Agile environment, implementing AES-256 encryption, RSA digital signatures, and SHA-512 hashing to guarantee end-to-end confidentiality and integrity across a client-server messaging system.

Technical Skills

Languages: Java, Python, Rust, C, TypeScript, JavaScript

Frameworks & Tools: FastAPI, Flask, React, Node.js, Docker, PostgreSQL, Redis, AWS, OpenCV